ThermoGuard AI Controled Safety System

The School of Engineering | Electrical & Electronics Engineering | Supervised by Dr. Mostafa Rabie

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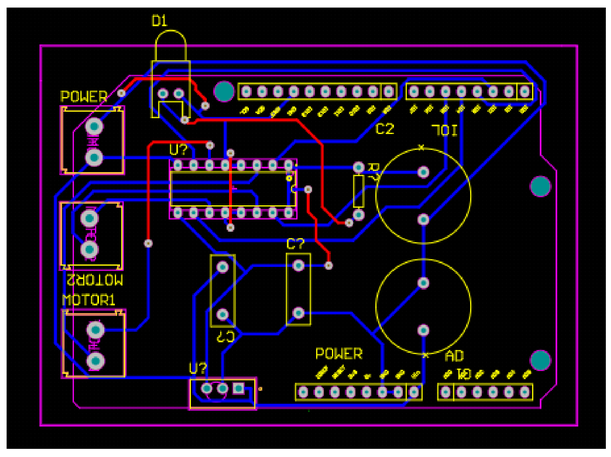
Introduction

ThermoGuard AC is an AI-enhanced automotive climate system designed and manufactured by us using a PCB for an Arduino motor Sheild to optimize passenger comfort and vehicle safety. It maintains temperatures between 16°C and 30°C using an intelligent feedback loop. The system features AI Vision to detect occupant counts and locations, allowing it to scale cooling power and target airflow directly toward people. Integrated Voice Interaction provides real-time status updates, while Solar Panels prioritize renewable energy to power the fan and electronics. For safety, the system monitors for overheating, triggering a full lockdown and voice alert if temperatures exceed 50°C.

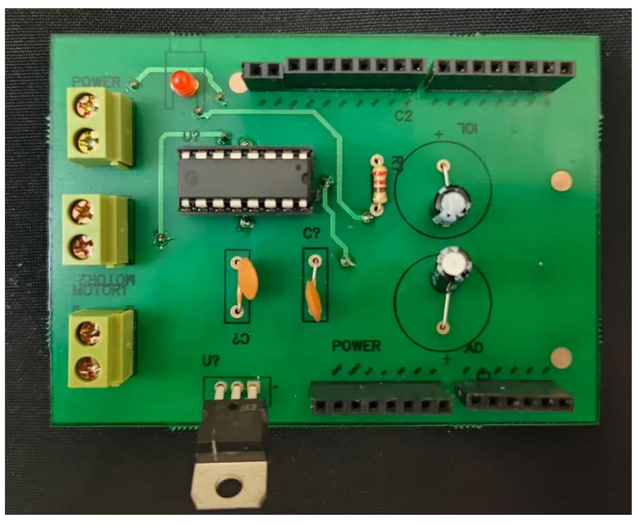
Methods

- Custom Hardware: Designed a 2-layer Arduino Motor Shield PCB optimized for L298N high-current routing.
- AI Sensing: ESP32-CAM module performs real-time occupancy counting and spatial coordinate tracking
- Targeted Flow: SG90 Servo uses AI data to aim air vents directly at detected occupant centroids.
- Interface: DFPlayer Mini provides interactive voice feedback and status alerts for all system modes.
- Green Power: Solar PV Panel and controller prioritize renewable energy to drive the fan and electronics

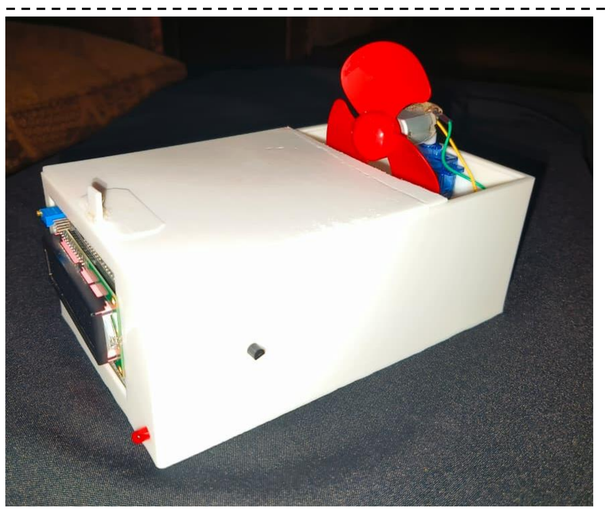
Layout PCB



Printed PCB



Final Design



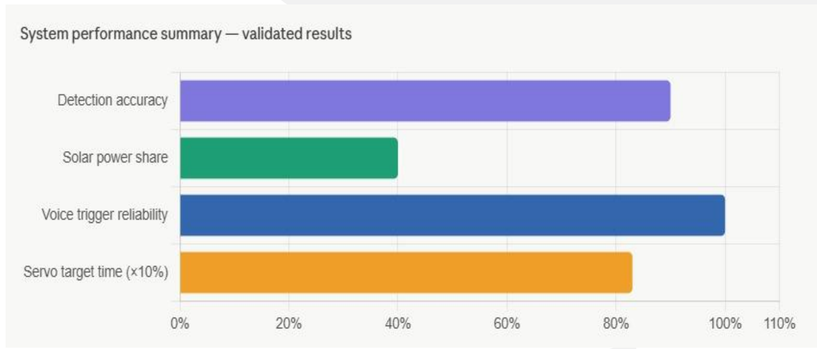
Data Analysis

The system was validated using test cases simulating all major operational scenarios to evaluate intelligent responsiveness. Key behaviors analyzed include:

- **Proportional Control:** Fan speed scales precisely with thermal error: **100 PWM** (1–2°C), **170 PWM** (3–4°C), and **255 PWM** (≥ 5°C)
- **Turbo Mode:** KEY_RIGHT forces **MAX (255 PWM)** and overrides the target to **16°C** regardless of current ambient temperature.
- **Safety Logic:** Shutdown triggers at **>50°C**; alarms activate and the fan runs at **100% for 5s** before a total system halt.
- **AI Occupancy Logic:** The fan duty cycle increases by **15% per person** detected by the ESP32-CAM to proactively adjust for metabolic heat gain.
- **Spatial Targeting:** Airflow direction is mapped to **X-coordinate AI data**, shifting the vent toward occupants within **1.2s** using precision servo positioning.
- **Energy Harvesting:** Monitoring confirmed that **Solar PV integration** provided up to **40% of operational power** during peak sunlight, reducing battery drain.
- **Voice Interaction:** The DFPlayer Mini achieved **100% trigger reliability**, providing interactive audio feedback for mode transitions and emergency safety warnings.







Results

The integration of **AI Vision**, **Voice Feedback**, and **Solar Power** into the **ThermoGuard** system yielded the following results during validation:

- **Detection Accuracy**: The **ESP32-CAM** successfully identified and counted occupants with 90% accuracy, triggering immediate fan speed adjustments.
- **Targeting Precision**: Airflow was directed to occupant centroids within **1.2 seconds** of movement detection, confirmed by mapping camera X-coordinates to the **SG90 Servo**.
- **Audio Reliability**: The **DFPlayer Mini** achieved 100% trigger reliability, replacing buzzer tones with clear voice prompts for "Turbo Mode," "Eco Mode," and "Emergency" status.
- **Energy Efficiency**: **Solar PV integration** provided up to **40% of the total power** required during peak daylight hours, significantly extending system runtime.
- **Safety Enforcement**: The system consistently executed a total lockdown and voiced an alarm when simulated temperatures exceeded **50°C**.
- **Hardware Integrity**: The customized **2-layer PCB (Arduino Motor Shield)** reliably managed high-current paths for the **L298N** without overheating.
- **User Interface**: The **16x2 LCD** provided clear real-time data on temperature and occupant counts, complemented by interactive voice status updates.

Conclusion

The ThermoGuard successfully demonstrates a complete embedded automotive AC system using Native C with direct register access. By integrating AI Vision, interactive Voice Feedback, and Solar Power, the project transforms from a reactive controller into an intelligent cabin environment manager. Utilizing a customized 2-layer PCB motor shield, the system achieved its core objectives:

- **Monitoring**: Real-time temperature is displayed on the LCD with custom animations.
- **Control**: Fan speed and direction automatically regulate cabin temperature proportionally, further enhanced by AI spatial targeting to aim vents directly at occupants.
- **AI & Sustainability**: AI occupancy sensing scales cooling power based on person count, while solar PV integration prioritizes renewable energy for the fan and electronics.
- **Safety & Efficiency**: Emergency shutdown at **>50°C** and **Eco Mode** during inactivity are fully verified with integrated voice alerts.

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